

CRISTIAN CAMILO PUENTES DIX

+57 3226124247

Medellín - Colombia

Email: rugal8546@gmail.com

LinkedIn: <https://www.linkedin.com/in/cpd2001>

Github: <https://github.com/cris185>

Portfolio: <https://cristianpuentes.com>

PROFESSIONAL SUMMARY

BACKEND ENGINEER | FULL STACK DEVELOPER (Hybrid) Systems and Computer Engineer who has shipped four production SaaS platforms end-to-end — spanning payment processing, subscription billing, appointment scheduling, and AI-powered automation. Backend specialist in Python (Django, Django REST Framework), Node.js, Express.js, NestJS, and SQL databases.

Currently working as a Full Stack Developer building production applications with React, Next.js, TypeScript, Node.js, SQL, and AWS. Experienced in designing REST APIs, payment integrations, subscription systems, messaging workflows, transactional processes, and scalable backend solutions.

Builds and ships independent full-stack products end-to-end — including a 25+ model telemedicine platform and an LSTM-based stock prediction engine — with hands-on machine learning experience from an AI-based candidate-matching thesis.

PROFESSIONAL EXPERIENCE

Geeks5g — Backend & Full Stack Developer

June 2025 - Present

Delivered and maintained four production SaaS platforms end-to-end, from database schema to deployed UI, across Next.js/React, Node.js/Express, NestJS, PostgreSQL, and Redis.

- **Verify Lead** — lead-verification and outbound email platform for a 40+ person sales team: built deliverability checks before a lead enters a campaign, credit-based usage limits, and scheduled email sends with delivery metrics; integrated PayArc for billing, SendGrid for email, and Twilio for SMS.
- **Dashboard Analytics** — full-stack scheduling and operations platform for a spa/wellness business: built conflict-aware appointment scheduling across staff and rooms, role-based access control, OCR-assisted intake forms, and an in-app wallet for tracking charges and refunds.
- **RankTitan.ai** — AI-powered local SEO platform: contributed street-by-street rank tracking via geo-grid heat maps, AI-drafted review-reply automation, and one-click Google Business Profile management; stack included NestJS, Next.js, Prisma/PostgreSQL, and the Gemini and Meta Graph APIs.
- **Platform Connection** — AI chatbot platform for customer-service automation: built multilingual conversation flows, appointment scheduling synced to Google/Microsoft calendars, and multi-channel messaging across WhatsApp and Meta.
- Designed and consumed RESTful APIs across decoupled frontend/backend and monorepo architectures; applied Atomic Design and Domain-Driven Design (DDD) on the frontend.

Universidad Tecnológica de Bolívar — IAI & Databases Researcher (Professional Internship)

January 2024 - July 2024

- Designed and prototyped an institutional chatbot using OpenAI APIs and automation tools.
- Implemented conversational AI solutions integrated with databases for real-time information queries.
- Developed logic based on function calling to transform user requests into dynamic SQL queries.
- Designed database models to optimize traceability and inventory control.
- Produced technical documentation covering the full research and development process: Python, SQL, OpenAI APIs, Postman, Power Automate.

PROJECTS

Personal Project – CHO Health

2026 - present

Designed and developed an end-to-end telemedicine platform supporting appointment scheduling, online payments, medical consultations, prescriptions, laboratory orders, invoicing, and pharmacy delivery workflows.

- Built a scheduling engine with validation of working hours, breaks, service duration, and appointment conflicts.
- Implemented three-layer booking protection using frontend validation, DRF serializer validation and database constraints.
- Integrated Stripe and PayPal payment processing with idempotent webhook handling.
- Developed refund workflows with business-rule enforcement and audit tracking.
- Implemented atomic medical consultation workflows combining diagnosis, prescriptions, and laboratory orders.
- Generated invoices, prescriptions, and laboratory orders as PDF documents using ReportLab.
- Integrated SendGrid transactional emails, Cloudinary media storage, QR pickup codes, and multilingual support.
- Designed and implemented 25+ database models, 70+ REST API endpoints, and 40+ frontend routes.

Technologies: Django, Django REST Framework, PostgreSQL, Next.js, React, TypeScript, Stripe, PayPal, SendGrid, ReportLab, Cloudinary, Supabase, Render.

Personal Project – Seven Lever

2026 - present

A local two-player dice game based on "Seven Eleven," the game played in Colombian schoolyards, with a full ruleset, betting, and an animated loss sequence.

- Built the complete game engine as pure, side-effect-free frontend logic (Next.js, TypeScript) for zero-latency turns – the backend only persists finished rounds and match history.
- Designed the data model (Player, Game, Round, Roll) and REST API (Express, Prisma, PostgreSQL) for match history and stats.
- Implemented Framer Motion animations for dice rolls and the loss sequence, with lazy-loaded sound effects via Howler.js.

Technologies: Next.js, TypeScript, Zustand, Express, Prisma, PostgreSQL, Framer Motion.

Personal Project – CHO Planner

2026 - present

A self-hosted, local-first task and note manager with a swappable AI assistant (Claude, GPT, or Gemini) that breaks goals into weighted subtasks.

- Built a Bring-Your-Own-Key architecture – each user supplies their own LLM and Google Calendar credentials, encrypted at rest with AES-256-GCM and never sent back to the frontend.
- Validated every AI provider's output against a shared Zod schema with automatic retries, so switching models never changes the structure of what gets generated.
- Built a Kanban board with drag-and-drop, weighted task priorities, sprints, Markdown notes, and optional Google Calendar sync via OAuth 2.0.
- Shipped as an installable PWA with an offline page and a service worker that never caches API routes.

Technologies: Next.js, TypeScript, Prisma, PostgreSQL, Auth.js, Zod, Google Calendar API.

Personal Project — Stock Prediction Model (Neurostock)

2025 - present

Predicts stock prices for any ticker with an LSTM neural network trained on historical data, with backtesting and confidence intervals instead of a single-number guess.

- Built a 2-layer LSTM (128 + 64 units) trained on up to 10 years of historical price data per ticker.
- Applied Monte Carlo Dropout (50 iterations) to turn a single prediction into a real confidence interval.
- Enforced strict train/test scaler separation to prevent data leakage — a mistake most tutorials skip.
- Built a backtesting dashboard reporting MSE, RMSE, and R² against real historical prices.

Technologies: Django, Django REST Framework, PostgreSQL, React, TypeScript, TensorFlow/Keras, JWT.

Bachelor’s Thesis — AI-Based Talent Selection System (AI-JobMatch)

February 2023 – November 2023

- Developed a matching system between candidates and job openings using cosine similarity and Euclidean distance metrics.
- Integrated OpenAI APIs for transcription and candidate information analysis.
- Optimized the model to improve accuracy and representativeness in recruitment processes.

Technologies: Python, Machine Learning, OpenAI APIs.

EDUCATION

Universidad Tecnológica de Bolívar

2019 - 2024

Bachelor’s Degree in Systems and Computer Engineering

TECHNICAL SKILLS

- **Frontend:** React, Next.js, TypeScript, JavaScript, HTML, CSS, Framer Motion, Zustand.
- **Backend:** Python, Django, Django REST Framework, Node.js, Express.js, NestJS, REST APIs
- **Databases:** PostgreSQL, SQL, MongoDB, Prisma ORM, Redis.
- **Cloud & DevOps:** AWS, Docker, Git, GitHub, GitHub Actions, Render, Supabase.
- **AI & Automation:** OpenAI APIs, Gemini API, Machine Learning, Neural Networks (LSTM), TensorFlow/Keras.
- **Integrations:** Stripe, PayPal, PayArc, Twilio, SendGrid, Google Calendar API, Meta Graph API.

REFERENCES

Alexander Nieve Romero

Phone

Software Engineer - General Manager at Gesnnova

+57 3226141645

Santiago Rivera Cano

Phone

Mid-Level Full Stack Developer - Precision Logistica

+ 57 3113172984

Elias Jiménez Diz

Phone

Mid-Level Full Stack Developer - Geeks5g

+ 57 3013281320

Owen Tovar Puello

Phone

IA OPS Engineer - Healthatom

+ 57 3017659144

Hernan Zuluaga Ricardo

Phone

IA Engineer | Fullstack Developer - Sisma Corporation

+ 57 3203290036